

FRED TALKS

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Highlights from my conversation about agentic engineering on Lenny's Podcast

I was a guest on Lenny Rachitsky's podcast, in a new episode titled [An AI state of the union: We've passed the inflection point, dark factories are coming, and automation timelines](#). It's available on [YouTube](#), [Spotify](#), and [Apple Podcasts](#). Here are my highlights from our conversation, with relevant links.

[An AI state of the union: We've passed the inflection point & dark factories are coming](#)[Lenny's Podcast](#)



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The November inflection point

4:19—The end result of these two labs throwing everything they had at making their models better at code is that in November we had what I call the [inflection point](#) where GPT 5.1 and Claude Opus 4.5 came along.

They were both incrementally better than the previous models, but in a way that crossed a threshold where previously the code would mostly work, but you had to pay very close attention to it. And suddenly we went from that to... almost all of the time it does what you told it to do, which makes all of the difference in the world.

Now you can spin up a coding agent and say, [build me a Mac application that does this thing](#), and you'll get something back which won't just be a buggy pile of rubbish that doesn't do anything.

Software engineers as bellwethers for other information workers

5:49—I can churn out 10,000 lines of code in a day. And most of it works. Is that good? Like, how do we get from most of it works to all of it works? There are so many new questions that we’re facing, which I think makes us a bellwether for other information workers.

Code is easier than almost every other problem that you pose these agents because code is obviously right or wrong—either it works or it doesn’t work. There might be a few subtle hidden bugs, but generally you can tell if the thing actually works.

If it writes you an essay, if it prepares a lawsuit for you, it’s so much harder to derive if it’s actually done a good job, and to figure out if it got things right or wrong. But it’s happening to us as software engineers. It came for us first.

And we’re figuring out, OK, what do our careers look like? How do we work as teams when part of what we did that used to take most of the time doesn’t take most of the time anymore? What does that look like? And it’s going to be very interesting seeing how this rolls out to other information work in the future.

Lawyers are falling for this really badly. The [AI hallucination cases database](#) is up to 1,228 cases now!

Plus this bit from the cold open at [the start](#):

It used to be you’d ask ChatGPT for some code, and it would spit out some code, and you’d have to run it and test it. The coding agents take that step for you now. And an open question for me is how many other knowledge work fields are actually prone to these agent loops?

Writing code on my phone

8:19—I write so much of my code on my phone. It’s wild. I can get good work done walking the dog along the beach, which is delightful.

I mainly use the Claude iPhone app for this, both with a regular Claude chat session (which [can execute code now](#)) or using it to control [Claude Code for web](#).

Responsible vibe coding

9:55 If you're vibe coding something for yourself, where the only person who gets hurt if it has bugs is you, go wild. That's completely fine. The moment you ship your vibe coding code for other people to use, where your bugs might actually harm somebody else, that's when you need to take a step back.

See also [When is it OK to vibe code?](#)

Dark Factories and StrongDM

12:49 The reason it's called the dark factory is there's this idea in factory automation that if your factory is so automated that you don't need any people there, you can turn the lights off. Like the machines can operate in complete darkness if you don't need people on the factory floor. What does that look like for software? [...]

So there's this policy that nobody writes any code: you cannot type code into a computer. And honestly, six months ago, I thought that was crazy. And today, probably 95% of the code that I produce, I didn't type myself. That world is practical already because the latest models are good enough that you can tell them to rename that variable and refactor and add this line there... and they'll just do it—it's faster than you typing on the keyboard yourself.

The next rule though, is nobody *reads* the code. And this is the thing which StrongDM started doing last year.

I wrote a lot more about [StrongDM's dark factory explorations](#) back in February.

The bottleneck has moved to testing

21:27—It used to be, you'd come up with a spec and you hand it to your engineering team. And three weeks later, if you're lucky, they'd come back with an implementation. And now that maybe takes three hours, depending on how well the coding agents are established for that kind of thing. So now what, right? Now, where else are the bottlenecks?

Anyone who's done any product work knows that your initial ideas are always wrong. What matters is proving them, and testing them.

We can test things so much faster now because we can build workable prototypes so much quicker. So there's an interesting thing I've been doing in my own work where any feature that I want to design, I'll often prototype three different ways it could work because that takes very little time.

I've always loved prototyping things, and prototyping is even more valuable now.

22:40—A UI prototype is free now. ChatGPT and Claude will just build you a very convincing UI for anything that you describe. And that's how you should be working. I think anyone who's doing product design and isn't vibe coding little prototypes is missing out on the most powerful boost that we get in that step.

But then what do you do? Given your three options that you have instead of one option, how do you prove to yourself which one of those is the best? I don't have a confident answer to that. I expect this is where the good old fashioned usability testing comes in.

More on prototyping later on:

46:35—Throughout my entire career, my superpower has been prototyping. I've been very quick at knocking out working prototypes of things. I'm the person who can show up at a meeting and say, look, here's how it could work. And that was kind of my unique selling point. And that's gone. Anyone can do what I could do.

This stuff is exhausting

26:25—I'm finding that using coding agents well is taking every inch of my 25 years of experience as a software engineer, and it is mentally exhausting. I can fire up four agents in parallel and have them work on four different problems. And by like 11 AM, I am wiped out for the day. [...]

There's a personal skill we have to learn in finding our new limits—what's a responsible way for us not to burn out.

I've talked to a lot of people who are losing sleep because they're like, my coding agents could be doing work for me. I'm just going to stay up an extra half hour and set off a bunch of extra things... and then waking up at four in the morning. That's obviously unsustainable. [...]

There's an element of sort of gambling and addiction to how we're using some of these tools.

Interruptions cost a lot less now

45:16—People talk about how important it is not to interrupt your coders. Your coders need to have solid two to four hour blocks of uninterrupted work so they can spin up their mental model and churn out the code. That's changed completely. My programming work, I need two minutes every now and then to prompt my agent about what to do next. And then I can do the other stuff and I can go back. I'm much more interruptible than I used to be.

My ability to estimate software is broken

28:19—I've got 25 years of experience in how long it takes to build something. And that's all completely gone—it doesn't work anymore because I can look at a problem and say that this is going to take two weeks, so it's not worth it. And now it's like... maybe it's going to take 20 minutes because the reason it would have taken two weeks was all of the sort of cruffy coding things that the AI is now covering for us.

I constantly throw tasks at AI that I don't think it'll be able to do because every now and then it does it. And when it doesn't do it, you learn, right? But when it *does* do something, especially something that the previous models couldn't do, that's actually cutting edge AI research.

And a related anecdote:

36:56—A lot of my friends have been talking about how they have this backlog of side projects, right? For the last 10, 15 years, they've got projects they never quite finished. And some of them are like, well, I've done them all now. Last couple of months, I just went through and every evening I'm like, let's take that project and finish it. And they almost feel a sort of sense of loss at the end where they're like, well, okay, my backlog's gone. Now what am I going to build?

It's tough for people in the middle

29:29—So ThoughtWorks, the big IT consultancy, did an offsite about a month ago, and they got a whole bunch of engineering VPs in from different companies to talk about this stuff. And one of the interesting theories they came up with is they think this stuff is really good for experienced engineers, like it amplifies their skills. It's really good for new engineers because it solves so many of those onboarding problems. The problem is

the people in the middle. If you're mid-career, if you haven't made it to sort of super senior engineer yet, but you're not sort of new either, that's the group which is probably in the most trouble right now.

I mentioned [Cloudflare hiring 1,000 interns](#), and Shopify too.

Lenny asked for my advice for people stuck in that middle:

31:21—That's a big responsibility you're putting on me there! I think the way forward is to lean into this stuff and figure out how do I help this make me better?

A lot of people worry about skill atrophy: if the AI is doing it for you, you're not learning anything. I think if you're worried about that, you push back at it. You have to be mindful about how you're applying the technology and think, okay, I've been given this thing that can answer any question and *often* gets it right. How can I use this to amplify my own skills, to learn new things, to take on much more ambitious projects? [...]

33:05—Everything is changing so fast right now. The only universal skill is being able to roll with the changes. That's the thing that we all need.

The term that comes up most in these conversations about how you can be great with AI is *agency*. I think agents have no agency at all. I would argue that the one thing AI can never have is agency because it doesn't have human motivations.

So I'd say that's the thing is to invest in your own agency and invest in how to use this technology to get better at what you do and to do new things.

It's harder to evaluate software

The fact that it's so easy to create software with detailed documentation and robust tests means it's harder to figure out what's a credible project.

37:47 Sometimes I'll have an idea for a piece of software, Python library or whatever, and I can knock it out in like an hour and get to a point where it's got documentation and tests and all of those things, and it looks like the kind of software that previously I'd have spent several weeks on—and I can stick it up on GitHub

And yet... I don't believe in it. And the reason I don't believe in it is that I got to rush through all of those things... I think the quality is probably good, but I haven't spent enough time with it to feel confident in that quality. Most importantly, *I haven't used it yet*.

It turns out when I'm using somebody else's software, the thing I care most about is I want them to have used it for months.

I've got some very cool software that I built that I've *never used*. It was quicker to build it than to actually try and use it!

The misconception that AI tools are easy

41:31—Everyone's like, oh, it must be easy. It's just a chat bot. It's not easy. That's one of the great misconceptions in AI is that using these tools effectively is easy. It takes a lot of practice and it takes a lot of trying things that didn't work and trying things that did work.

Coding agents are useful for security research now

19:04—In the past sort of three to six months, they've started being credible as security researchers, which is sending shockwaves through the security research industry.

See Thomas Ptacek: [Vulnerability Research Is Cooked](#).

At the same time, open source projects are being bombarded with junk security reports:

20:05—There are these people who don't know what they're doing, who are asking ChatGPT to find a security hole and then reporting it to the maintainer. And the report looks good. ChatGPT can produce a very well formatted report of a vulnerability. It's a total waste of time. It's not actually verified as being a real problem.

A good example of the right way to do this is [Anthropic's collaboration with Firefox](#), where Anthropic's security team *verified* every security problem before passing them to Mozilla.

OpenClaw

Of course we had to talk about OpenClaw! Lenny had his running on a Mac Mini.

1:29:23—OpenClaw demonstrates that people want a personal digital assistant so much that they are willing to not just overlook the security side of things, but also getting the thing running is not easy. You’ve got to create API keys and tokens and install stuff. It’s not trivial to get set up and hundreds of thousands of people got it set up. [...]

The first line of code for OpenClaw was written on November the 25th. And then in the Super Bowl, there was an ad for AI.com, which was effectively a vaporware white labeled OpenClaw hosting provider. So we went from first line of code in November to Super Bowl ad in what? Three and a half months.

I continue to love Drew Breunig’s description of OpenClaw as a digital pet:

A friend of mine said that OpenClaw is basically a Tamagotchi. It’s a digital pet and you buy the Mac Mini as an aquarium.

Journalists are good at dealing with unreliable sources

In talking about my explorations of AI for data journalism through Datasette:

1:34:58—You would have thought that AI is a very bad fit for journalism where the whole idea is to find the truth. But the flip side is journalists deal with untrustworthy sources all the time. The art of journalism is you talk to a bunch of people and some of them lie to you and you figure out what’s true. So as long as the journalist treats the AI as yet another unreliable source, they’re actually better equipped to work with AI than most other professions are.

The pelican benchmark

Of course we talked about pelicans riding bicycles:

56:10—There appears to be a very strong correlation between how good their drawing of a pelican riding a bicycle is and how good they are at everything else. And nobody can explain to me why that is. [...]

People kept on asking me, what if labs cheat on the benchmark? And my answer has always been, really, all I want from life is a really good picture of a pelican riding a bicycle. And if I can trick every AI lab in the world into cheating on benchmarks to get it, then that just achieves my goal.

59:56—I think something people often miss is that this space is inherently funny. The fact that we have these incredibly expensive, power hungry, supposedly the most advanced computers of all time. And if you ask them to draw a pelican on a bicycle, it looks like a five-year-old drew it. That's really funny to me.

And finally, some good news about parrots

Lenny asked if I had anything else I wanted to leave listeners with to wrap up the show, so I went with the best piece of news in the world right now.

1:38:10—There is a rare parrot in New Zealand called the Kākāpō. There are only 250 of these parrots left in the world. They are flightless nocturnal parrots—beautiful green dumpy looking things. And the good news is they're having a fantastic breeding season in 2026,

They only breed when the Rimu trees in New Zealand have a mass fruiting season, and the Rimu trees haven't done that since 2022—so there has not been a single baby kākāpō born in four years.

This year, the Rimu trees are in fruit. The kākāpō are breeding. There have been dozens of new chicks born. It's a really, really good time. It's great news for rare New Zealand parrots and you should look them up because they're delightful.

Everyone should [watch the live stream of Rakiura on her nest with two chicks!](#)

YouTube chapters

Here's the full list of chapters Lenny's team defined for the YouTube video:

- 00:00: Introduction to Simon Willison
- 02:40: The November 2025 inflection point
- 08:01: What's possible now with AI coding
- 10:42: Vibe coding vs. agentic engineering
- 13:57: The dark-factory pattern
- 20:41: Where bottlenecks have shifted

- [23:36](#): Where human brains will continue to be valuable
- [25:32](#): Defending of software engineers
- [29:12](#): Why experienced engineers get better results
- [30:48](#): Advice for avoiding the permanent underclass
- [33:52](#): Leaning into AI to amplify your skills
- [35:12](#): Why Simon says he's working harder than ever
- [37:23](#): The market for pre-2022 human-written code
- [40:01](#): Prediction: 50% of engineers writing 95% AI code by the end of 2026
- [44:34](#): The impact of cheap code
- [48:27](#): Simon's AI stack
- [54:08](#): Using AI for research
- [55:12](#): The pelican-riding-a-bicycle benchmark
- [59:01](#): The inherent ridiculousness of AI
- [1:00:52](#): Hoarding things you know how to do
- [1:08:21](#): Red/green TDD pattern for better AI code
- [1:14:43](#): Starting projects with good templates
- [1:16:31](#): The lethal trifecta and prompt injection
- [1:21:53](#): Why 97% effectiveness is a failing grade
- [1:25:19](#): The normalization of deviance
- [1:28:32](#): OpenClaw: the security nightmare everyone is looking past
- [1:34:22](#): What's next for Simon
- [1:36:47](#): Zero-deliverable consulting

- [1:38:05](#): Good news about Kakapo parrots

A non-anthropomorphized view of LLMs

HALVAR.FLAKE · 14 SEP 2025 · [SOURCE](#)

In many discussions where questions of "alignment" or "AI safety" crop up, I am baffled by seriously intelligent people imbuing almost magical human-like powers to something that - in my mind - is just MatMul with interspersed nonlinearities.

In one of these discussions, somebody correctly called me out on the simplistic nature of this argument - "a brain is just some proteins and currents". I felt like I should explain my argument a bit more, because it feels less simplistic to me:

The space of words

The tokenization and embedding step maps individual words (or tokens) to some vectors. So let us imagine for a second that we have in front of us. A piece of text is then a path through this space - going from word to word to word, tracing a (possibly convoluted) line.

Imagine now that you label each of the "words" that form the path with a number: The last word with 1, counting forward until you hit the first word or the maximum context length . If you've ever played the game "Snake", picture something similar, but played in very high-dimensional space - you're moving forward through space with the tail getting truncated off.

The LLM takes your previous path into account, calculates probabilities for the next point to go to, and then makes a random pick into the next point according to these probabilities. An LLM instantiated with a fixed random seed is a mapping of the form .

In my mind, the paths generated by these mappings look a lot like [strange attractors](#) in dynamical systems - complicated, convoluted paths that are structured-ish.

Learning the mapping

We obtain this mapping by training it to mimic human text. For this, we use approximately all human writing we can obtain, plus corpora written by human experts on a particular topic, plus some automatically generated pieces of text in domains where we can automatically generate and validate them.

Paths to avoid

There are certain language sequences we wish to avoid - because the sequences these models generate try to mimic human speech in all its empirical structure, but we feel that some of the things that humans have empirically written are very undesirable to be generated. We also feel that a variety of other paths should ideally not be generated, if - when interpreted by either humans or other computer systems - undesirable results arise.

We can't specify strictly in a mathematical sense which paths we would prefer not to generate, but we can provide examples and counterexamples, and we try to hence nudge the complicated learnt distribution away from them.

"Alignment" for LLMs

Alignment and safety for LLMs mean that **we should be able to quantify and bound the probability with which certain undesirable sequences are generated**. The trouble is that we largely fail at describing "undesirable" except by example, which makes calculating bounds difficult.

For a given LLM (without random seed) and sequence, it is trivial to calculate the probability of the sequence to be generated. So if we had a way of somehow summing or integrating over these probabilities, we could say with certainty "this model will generate an undesirable sequence once every N model evaluations". We can't, currently, and that sucks, but at the heart, this is the mathematical and computational problem we'd need to solve.

The surprising utility of LLMs

LLMs solve a large number of problems that could previously not be solved algorithmically. NLP (as the field was a few years ago) has largely been solved.

I can write a request in plain English to summarize a document for me and put some key datapoints from the document in a structured JSON format, and modern models will just do that. I can ask a model to generate a children's book story involving raceboats and generate illustrations, and the model will generate something that is passable. And much more, all of which would have seemed like absolute science fiction 5-6 years ago.

We're on a pretty steep improvement curve, so I expect the number of currently-intractable problems that these models can solve to keep increasing for a while.

Where anthropomorphization loses me

The moment that people ascribe properties such as "consciousness" or "ethics" or "values" or "morals" to these learnt mappings is where I tend to get lost. We are speaking about a big recurrence equation that produces a new word, and that stops producing words if we don't crank the shaft.

To me, wondering if this contraption will "wake up" is similarly bewildering as if I was to ask a computational meteorologist if he isn't afraid of his meteorological numerical calculation will "wake up".

I am baffled that the AI discussions seem to never move away from treating a function to generate sequences of words as something that resembles a human. Statements such as "an AI agent could become an insider threat so it needs monitoring" are simultaneously unsurprising (you have a randomized sequence generator fed into your shell, literally **anything** can happen!) and baffling (you talk as if you believe the dice you play with had a mind of their own and could decide to conspire against you).

Instead of saying "we cannot ensure that no harmful sequences will be generated by our function, partially because we don't know how to specify and enumerate harmful sequences", we talk about "behaviors", "ethical constraints", and "harmful actions in pursuit of their goals". All of these are anthropocentric concepts that - in my mind - do not apply to functions or other mathematical objects. And using them muddles the discussion, and our thinking about what we're doing when we create, analyze, deploy and monitor LLMs.

This muddles the public discussion. We have many historical examples of humanity ascribing bad random events to "the wrath of god(s)" (earthquakes, famines, etc.), "evil spirits" and so forth. The fact that intelligent highly educated researchers talk about these mathematical objects in anthropomorphic terms makes the technology seem mysterious, scary, and magical.

We should think in terms of "this is a function to generate sequences" and "by providing prefixes we can steer the sequence generation around in the space of words and change the probabilities for output sequences". And for every possible undesirable output sequence of a length smaller than , we can pick a context that maximizes the probability of this undesirable output sequence.

A much clearer formulation, which helps more clearly articulate the problems to solve.

Why many AI luminaries tend to anthropomorphize

Perhaps I am fighting windmills, or rather a self-selection bias: A fair number of current AI luminaries have self-selected by their belief that they might be the ones getting to AGI - "creating a god" so to speak, the creation of something like life, as good as or better than humans. You are more likely to choose this career path if you believe that it is feasible, and that current approaches might get you there. Possibly I am asking people to "please let go of the belief that you based your life around" when I am asking for an end to anthropomorphization of LLMs, which won't fly.

Why I think human consciousness isn't comparable to an LLM

The following is uncomfortably philosophical, but: In my worldview, humans are dramatically different things than a function . For hundreds of millions of years, nature generated new versions, and only a small number of these versions survived. Human thought is a poorly-understood process, involving enormously many neurons, extremely high-bandwidth input, an extremely complicated cocktail of hormones, constant monitoring of energy levels, and millions of years of harsh selection pressure.

We understand essentially nothing about it. In contrast to an LLM, given a human and a sequence of words, I cannot begin putting a probability on "will this human generate this sequence".

To repeat myself: To me, considering that any human concept such as ethics, will to survive, or fear, apply to an LLM appears similarly strange as if we were discussing the feelings of a numerical meteorology simulation.

The real issues

The function class represented by modern LLMs are very useful. Even if we never get anywhere close to AGI and just deploy the current state of technology everywhere where it might be useful, we will get a dramatically different world. LLMs might end up being similarly impactful as electrification.

My grandfather lived from 1904 to 1981, a period which encompassed moving from gas lamps to electric, the replacement of horse carriages by cars, nuclear power, transistors, all the way to computers. It also spanned two world wars, the rise of Communism and Stalinism, almost the entire lifetime of the USSR and GDR etc. The world on his birth looked nothing like the world when he died.

Navigating the dramatic changes of the next few decades while trying to avoid world wars and murderous ideologies is difficult enough without muddying our thinking.

Ownership

THORSTEN BALL · 08 JUL 2026 · [SOURCE](#)

Thoughts on ownership I sent to the Amp team in internal note

The following is an internal Slack message I sent to my Amp teammates after I had a conversation with one of them about ownership. It's only lightly edited.

I shared it before, but not here, because I didn't think too much of it. Then today, someone said their CTO shared my post with them and I thought: well, now I have to put it in the newsletter, don't I? So here we are.

Below, after the Slack post, I added some thoughts on juniors on how I see this advice applying to them.



Just had a (great) conversation about ownership and engineering here and I realized that I often use the phrase “ownership” or allude to it, but haven’t explained what “ownership” means to me in a while.

So, ownership.

If I ask you “can you own this?” or “can you take care of this?” or “are you on it?” — what I’m doing is I’m asking you to *own* it, to own the solution of a problem from end to end. From “we have a problem” to “we don’t have to think about it again.”

That means, when you say that you’re owning something, the expectation is that you...

- Think about what the problem actually is. Maybe you already have a solution in mind, without having thought about what we’re actually trying to solve here. Maybe you think “the problem is that we need to migrate from using X to using Y”, but that’s not a problem, that’s a solution. The problem is likely something like “performance is bad”, “it’s not stable”, “it fails for customer x”. Maybe there’s other possible solutions to that? Think about those. What are the tradeoffs? What’s the best solution to go with considering these tradeoffs?
- Think about edge cases. What are they? Which ones are important? Which ones can we ignore?
- Think about failures. Network failures are a given, for example. How do we handle them? Retry? Well, how often? How long?
- Think about data flow. How much data is involved here? Does data need to be migrated? Cleaned up? How can I get my hands on data to properly test this? What invariants are in the data? What assumptions do I have about the shape of the data that I haven’t confirmed yet?
- Think about how you’d test this. How can I know that what I built is correct or not? Are tests enough? Do I need to manually poke at things? Is the difference visible on a screenshot or in a video?
- How would we announce this? How do we communicate it? Can you picture it? How does it fit into the larger picture of our roadmap? Questions or concerns in that area — push back! ask!
- Do the work, with precision, with care, with a sense of urgency, with calmness. Do not half-ass things. Before you merge, ask yourself: am I proud of this? would I show this to John Carmack and say “here’s what I built, under these constraints, with these tradeoffs?”
- Test it manually. Yes, there’s automated tests. But in 99% of cases you can manually test or confirm that what you built works: you can run it yourself, you can ask an agent to run through test scenarios, you can poke at the data before and after, you can take screenshots, you can make a demo. Are you sure that what you did actually solves the problem?

- Make sure it lands in production and *works in production*. Is it deployed? Did the deploy fail? Do you need to activate a feature flag? Does the feature flag work? Can you use it in production? Can you confirm it's actually deployed?
- If you think your colleagues need to know about this change, because it's a new feature they should all test, or it's a new convention in codebase, or maybe it's a tricky thing everybody needs to be aware of, or something else: let them know! Do not underestimate peripheral vision: knowing that person X yesterday changed the behavior of how Z works might save person Y three hours of debugging today when a bug report related to Z comes in.
- Do customers need to know? Who reported the bug? Who's blocked? Let them know.
- Does the world need to know? Announce that it's out.
- Are there follow-ups? Do you need to check on what you shipped in the logs? A week later maybe?

Yes, that's a lot. And there's actually more, because I'm sure I forgot some stuff.

But that's how you build a product in a small team. We don't have PMs, we don't have a Q&A department. We're small, but we're *great*, we can do all of that.

And it's always okay to ask for help, it's okay to ask questions, it's okay to redo things and triple-check. What's not okay is to implicitly assume that someone else will do the things here that you haven't thought about.



“How does this apply juniors? You can’t expect them to really do all of that?”

I've been asked these questions, or variations of them, after I shared the thoughts above and here's my answer.

I do *not* expect juniors to *do* all of these things right away. But I would expect them to read through the list and *aspire* to one day be able to do all of these things. Until then, they can and should ask for help.

In fact, I don't expect *anybody* to *always* do *all of these things* for *everything*. It's a mental checklist of things to consider — problem, edge cases, tradeoffs, deployment, customers, messaging, ... — but for quite a few things there aren't edge cases to consider. Or big tradeoffs to weigh. Or deployment is a solved problem. And maybe someone else does the messaging for you.

And I imagine that most of these things you shouldn't even consider when you work in a company with, say, 5000 employees. I've never worked in a company that large, only startups, so I can't speak to how to successfully ship a software feature at Apple, end to end.

But when you work in a small company in which there's only a single department, when you want to build things you're proud of, when you work with me and you say own something, I expect you to keep these things in mind and run through them before you declare something as done.



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